

GURDHARAM JEET SINGH

Full-Stack Software Engineer • Web Systems & Cloud Automation Architect

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer and Autonomous Cloud Automation Architect with proven production experience shipping modern React, Next.js, and TypeScript web applications and high-throughput backend microservices. Architect of a 5-channel 24/7 cloud media generation fleet on GitHub Actions (yt-auto) with dual-gate multimodal LLM judge verification, telephonic Voice AI speech agents over Exotel WebSockets, SIP, and VoIP (<700ms turn latency), dynamic private API reverse engineering suites (Frida ARM64 libc hooking, Ghidra), and high-throughput Redis/BullMQ distributed message queues processing 25,000+ daily events. Creator of commercial web applications including SOLUM Minerals, Branders, and Dream Heights.

TECHNICAL SKILLS

Frontend & Web Systems: React, Next.js (App Router, SSR, RSC), TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, WebSockets, Three.js, WebGL2 (Camera Kit), GLSL Shaders, Lenis Smooth Scroll, Technical SEO, GEO, JSON-LD Schema.

Backend & Distributed Architecture: Node.js, Express, Python (FastAPI, Asyncio), Go, PostgreSQL, Redis & BullMQ Queues, RESTful APIs, Webhook Orchestration, Docker, Linux CLI / Bash, systemd, Tailscale, Headless Playwright / Puppeteer.

Cloud, DevOps & AI Workflows: GitHub Actions CI/CD, AWS (S3, Lambda), GCP, Railway, Sarvam AI Indic LLMs, Groq LPU, Voice AI, SIP, VoIP, Silero VAD, RAG, Model Context Protocol (MCP), FFmpeg Audio/Video Automation, TensorFlow Lite INT8, Flutter Isolates.

Security & Reverse Engineering: Frida ARM64 Libc Hooking (libcrypto.so), Ghidra Binary Decompilation, XLua, IL2CPP, OWASP Top 10, Burp Suite, Network Protocol Decryption, Session Extraction, Private API Reconstruction.

KEY ENGINEERING PROJECTS

Autonomous 5-Channel YouTube Media Fleet (yt-auto) — Cloud / GitHub Actions | 2025 – Present

Lead Automation Architect • Open Source (github.com/gurination1/yt-auto)

- **100% Headless Cloud Fleet Execution:** Engineered resilient 5-channel media generation fleet running autonomously on GitHub Actions across 15 daily slots with zero local machine dependencies.
- **Dual-Gate Multimodal Quality Control:** Integrated FFmpeg black-frame detector (`blackdetect=d=0.8`) and Gemini Multimodal Judge AI (evaluating narrative flow, pacing, and subtitle sync) enforcing a strict $\geq 85/100$ publication threshold.
- **Algorithmic Audio & Video Rendering:** Built headless audio ducking and dynamic background scoring pipeline in Python/FFmpeg with millisecond word-level subtitle alignment (.ass dynamic styling).

SOLUM Minerals, Branders & Dream Heights Web Suite — Remote / Production | 2025 – Present

Full-Stack Web Architect & Systems Developer

- **SOLUM Minerals Platform:** Developed low-carbon capsule fertilizer & mineral geology web experience featuring interactive product showcase, responsive data displays, and dynamic visual storytelling with 99+ Lighthouse performance.
- **Branders Digital Experience Platform:** Engineered high-performance brand agency web platform with modern interactive components, fluid micro-interactions, and 120fps physics-hardened mobile navigation.
- **Dream Heights Real Estate Platform:** Architected real estate portal with spatial property filtering, virtual floor plan previews, interactive image carousels, and automated client inquiry pipelines.
- **Mobile & Edge Systems:** Built Fasal Doctor (plant pathology Flutter / TensorFlow Lite INT8) and DoodHisaab (dairy ledger system with Flutter Isolates).

Genesis Coding Agent & AST Self-Healing Engine — Autonomous Systems / Python | 2025 – Present

Lead Architect • AST Code Synthesis & Cloud Workflows

- **Autonomous AST Mutation Loops:** Engineered self-healing agent engine that parses Python Abstract Syntax Trees (AST), generates surgical code diffs, and validates test suites with zero manual intervention.
- **Playwright & Headless Cloud Integration:** Implemented end-to-end browser verification harnesses, capturing multi-surface DOM state and generating automated visual validation logs.

WORK EXPERIENCE

Gurdharam (Independent Software Consultancy) — Remote / Punjab, India | 2025 – Present

Full-Stack Web & Autonomous Systems Engineer

- **Telephonic Voice AI & Speech Agent Architecture:** Architected real-time telephony voice agent over Exotel WebSockets, SIP, and VoIP, achieving <700ms turn latency across Sarvam AI Indic models and Groq LPUs with Silero VAD barge-in for Neo and Priya conversational personae.
- **High-Throughput Webhook & Messaging Gateway:** Engineered automated outreach and webhook triage gateway using Official WhatsApp APIs, Redis, and BullMQ, reliably processing 25,000+ incoming webhooks with sub-5-minute SLA.
- **Web AR Lens Pipeline (SnapAR Studio):** Developed WebGL2 Camera Kit testing portal integrating custom GLSL shaders, 3D asset conversion pipelines, and automated package verification.

EDUCATION

Baba Farid Group of Institutions (BFGI) — Punjab, India | Expected Graduation: May 2027

Bachelor of Science in Computer Science & Software Systems

- **Core Curriculum:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Software Engineering.

CERTIFICATIONS & REGISTRATIONS

- **Govt. MSME Registered Enterprise:** Ministry of Micro, Small and Medium Enterprises (UDYAM-PB-03-0079744).
- **Developer Partnerships & Verification:** Sarvam AI Startup Developer Member (Priority Indic LLM rates) • Meta Verified business entity.